FLAIR: A Federated Benchmark and Baseline Repository for ICU Prediction Across 12+ Academic Medical Centers Using the Common Longitudinal ICU Format

Manuscript outline — NEJM AI Datasets, Benchmarks, and Protocols

CLIF Consortium
FLAIR Working Group

June 5, 2026

[Working outline. Bulleted content cues sit under every section so drafting prose is paint-by-numbers. Hard envelope: 3000 main-text words, 5 figures+tables combined, 40 references, structured abstract.]

Front matter

- **Title (working):** FLAIR: A Federated Benchmark for ICU Prediction Across 12+ Academic Medical Centers Using the Common Longitudinal ICU Format.
- **Short title** ($\leq$ 50 char): FLAIR: Federated ICU Benchmark.
- **Authors / affiliations:** **TODO:** populate; up to 10 authors; flag equal-contribution and co-senior authorship in a title-page footnote. ORCIDs required.
- **Corresponding author:** **TODO:** name, email, institution.
- **Funding:** **TODO:** grant numbers; restated as a "Funded by" line at the end of the abstract.
- **Conflicts of interest:** **TODO:** ICMJE + Convey forms within 12 months for every author.
- **AI-use disclosure:** **TODO:** name LLM(s); state which sections or artifacts they helped produce. Restate in cover letter.
- **Word / figure / table / reference counts:** Main text *[$\leq$ 3000]*; figures+tables *[$\leq$ 5]*; references *[$\leq$ 40]*.

Structured abstract *[≈ 250 words]*

Background

- ICU prediction models trained on public datasets (MIMIC, eICU, HiRID) routinely fail to generalize beyond the institutions that produced their training data.
- Patient-level data cannot be centralized under HIPAA / IRB constraints, so multi-institutional external validation requires a federated evaluation framework.
- No prior ICU benchmark combines a shared data schema, auditable per-task code, aggregate-only outputs, and TRIPOD-AI reporting.

Methods

- Federated benchmark built atop CLIF v2.1 across N CLIF-consortium sites *[fill N at submission]*.
- Five clinically grounded tasks: T1 ICU daily mortality, T2 ICU daily LTACH discharge, T3 extubation failure <24h, T4 sepsis (antibiotic-anchored) <6h, T5 AKI KDIGO-2 <24h.
- Reference baseline kit: (i) a generic CLIF event tokenizer that maps every clinical event into a token and emits per-window count features, and (ii) two reference models — L2-penalized logistic regression and XGBoost — trained on MIMIC-CLIF and shipped inside the CLIF-WorkBench container for federated evaluation. Sites emit four aggregated JSON reports (`table1`, `tripod_ai`, `fairness`, `curves`).
- Open-source release (Apache 2.0, `github.com/clif-consortium/FLAIR`); the five-task suite is a seed — the catalog grows by community PR, each new task added as a single Python module under `flair_benchmark/tasks/`.

Results

- Combined cohort: **TODO:** n hospitalizations, n ICU stays, n sites across the consortium. MIMIC-CLIF baseline cohort (development entry point): 84,854 ICU stays at the broadest task (T4 sepsis); 26,771–65,577 stays for the four task-specific cohorts (T1/T2 daily disposition, T3 extubation, T5 AKI); per-task positive-rate range 1.7%–19.2% at decision-row granularity (see Table 1).
- Per-task AUROC / AUPRC (median, IQR across sites) **TODO:** fill from federated runs; calibration drift quantified between MIMIC-CLIF and consortium sites.
- Fairness deltas across sex, race, ethnicity, and age bins; bit- identical SHA-256 hashes across sites confirm protocol fidelity.

Conclusions

- FLAIR enables external validation of ICU AI models without data sharing. Code, container, and per-task schematics released openly. (Funded by **TODO:** ....)

1 Introduction *[≈ 600 words]*

- **P1 — Clinical stakes.** ICU decisions are real-time and high-consequence; rich EHR signal is available but underused.
- **P2 — The saturation problem in public benchmarks.** YAIB (van de Water et al., 2023) and HiRID-ICU-Benchmark (Yèche et al., 2021) harmonize public datasets (MIMIC, eICU, HiRID, AUMCdb); strong internal performance does not imply external generalization.
- **P3 — The privacy barrier.** HIPAA / IRB rules block centralization of PHI; multi-site validation must therefore be federated.
- **P4 — Why prior federated ICU efforts fall short.** Most rely on ad-hoc schemas, lack auditable per-task code, and do not emit TRIPOD-AI-aligned reports.
- **P5 — FLAIR's contribution.** Five clinically grounded tasks; CLIF v2.1 schema-locked code; single-file polars task scripts stamped with SHA-256; **a generic event-tokenizer ETL plus two reference baselines (logistic regression, XGBoost) released as the FLAIR baseline kit**; aggregate-only report bundle; baseline + federated results across the CLIF consortium.
- **P5b — Open source, extensible.** Apache 2.0 release at `github.com/clif-consortium/FLAIR`; the five-task suite is a seed catalog, with new tasks added by community PR as single Python modules under `flair_benchmark/tasks/` and accepted via two-site governance review.
- **Figure 1 callout** *[counts toward 5-item budget]*: *Summary of Contributions* panel modeled on EchoNext-Mini Fig. 1 (Hughes et al., 2026) — one side shows the federated workflow (MIMIC-CLIF → event tokenizer → count features → {logistic regression, XGBoost} → frozen baseline kit (Singularity) → consortium sites → aggregate JSONs); the other shows the five-task taxonomy.

2 Methods *[≈ 1200 words — heaviest section]*

Dataset curation

- CLIF v2.1 data standard maintained by 12+ US academic medical centers; combined coverage ∼2,000 ICU beds nationwide.
- MIMIC-CLIF as public development entry point (∼70k ICU stays; PhysioNet); identical schema guarantees code runs unmodified on consortium data.
- Site enrollment and IRB framework: each site IRB-approved independently; central protocol coordination at the University of Chicago **TODO:** confirm.
- CLIF tables exposed by `flair_benchmark/_clif.py`: `hospitalization`, `adt`, `patient`, `vitals`, `labs`, `respiratory_support`, `medication_admin_intermittent`, `medication_admin_continuous`.
- Data-quality gates inherited from `clifpy` validators; timezone handling per site `clif_config.json`.

Task definitions

Taxonomy. Each task is declared on three axes in its `META` dict: *task class* (outcome / deterioration / intervention), *temporal pattern* (episode / continuous), and *prediction unit* (one-per-stay / one-per-timestep). Full algorithmic specs sit in `docs/task_schematics/` and are included by reference in the Supplement.

- **T1 — ICU Daily In-Hospital Mortality (7 AM).** one-per-timestep, 24 h lookback, daily 07:00 local anchor. *Cohort:* all adult first-ICU-stay hospitalizations (no `discharge_category` filter). *Label:* $y = 1$ iff CLIF `discharge_category` $\in$ {`Expired`, `Hospice`}; all other dispositions (`Home`, `SNF`, `Rehab`, `Long Term Care Hospital (LTACH)`, `AMA`, `Other`, `Still Admitted`, null, ...) are label-0. *MIMIC-CLIF:* 64,434 stays / 198,125 daily rows; per-row positive rate 19.2%; stays-ever-positive 13.0%.
- **T2 — ICU Daily LTACH Discharge (7 AM).** Same cohort and grid as T1. *Label:* $y = 1$ iff CLIF `discharge_category` = `Long Term Care Hospital (LTACH)`; everything else — `Expired`, `Hospice`, `Home`, `SNF`, `Rehab`, `AMA`, `Other`, `Still Admitted`, null — is label-0. T1 and T2 do **not** partition: a Home discharge is 0 in both; an Expired discharge is 1 in T1 / 0 in T2. *MIMIC-CLIF:* same 64,434 stays / 198,125 daily rows; per-row positive rate 14.3%; stays-ever-positive 6.4% (LTACH-bound stays are long, so the per-row rate exceeds the stay-level rate).
- **T3 — Extubation Failure within 24 h.** one-per-stay, anchored at the first confirmed extubation event; consensus-window IMV episode detector in `_ventilation_episodes.py`. Label = death OR reintubation within 24 h post-extubation. *MIMIC-CLIF:* 26,771 extubation episodes; prevalence 5.4% (1,456 / 26,771).
- **T4 — Sepsis (antibiotic-anchored) within 6 h.** one-per-timestep, hourly grid; t_{zero} = first qualifying systemic IV antibiotic; label-leakage gate drops decision points $\geq t_{\text{zero}}$ (Kamran et al., 2024). *MIMIC-CLIF:* 84,854 stays / 7,093,633 hourly rows; per-row positive rate 3.3%; stays-ever-positive 54.9%.
- **T5 — AKI KDIGO-2 within 24 h.** one-per-timestep, hourly grid; SCr-only criterion; Koyner-2018 exclusions (Koyner et al., 2018). *MIMIC-CLIF:* 65,577 stays / 4,135,079 hourly rows; per-row positive rate 1.7%; stays-ever-positive 4.8%.

MIMIC-CLIF cohort snapshot. The numbers below come from `flair extract-metadata -clif-config clif_config_mimic.json` (`flair_benchmark` v2.0.0; HIPAA cell-suppression at $n<10$; SOFA computed as the worst score across the full hospitalization via `clifpy`'s polars backend (`compute_sofa_polars`); Charlson and Elixhauser computed from `HospitalDiagnosis` where available — *note: 49–54% of stays are missing comorbidity scores in the MIMIC-CLIF export, a known data-quality gap to flag in the Limitations*).

[This table is a drafting hook for Results—Table 1 and Methods—Cohort. It is not part of the 5-item NEJM AI figure/table budget; the submission Table 1 will mirror this layout across consortium sites with site-level median/IQR.]

Splits. Temporal date-based train/test split; MIMIC fallback 70/30 when split dates not specified.

	T1 mort.	T2 LTACH	T3 ext-fail	T4 sepsis	T5 AKI
ICU stays	64,434	64,434	26,771	84,854	65,577
Decision rows	198,125	198,125	26,771	7,093,633	4,135,079
Positive rate (row)	19.2%	14.3%	5.4%	3.3%	1.7%
Stays ever positive	13.0%	6.4%	5.4%	54.9%	4.8%
Sex: F / M	27,805/36,629	27,805/36,629	10,033/16,738	37,552/47,302	28,867/36,710
Age 65+ share	55.3%	55.3%	54.1%	54.6%	55.1%
SOFA hosp. max (med., IQR)	5 (2–8)	5 (2–8)	7 (5–10)	5 (2–8)	4 (2–7)
Charlson (median, IQR)	2 (0–4)	2 (0–4)	2 (0–4)	2 (0–4)	2 (0–4)

Table 1: MIMIC-CLIF cohort snapshot per task, generated by `flair extract-metadata`. *Decision rows* = one-per-stay for T3 (extubation episode), one-per-timestep otherwise (daily at 07:00 for T1/T2; hourly for T4/T5). *Age 65+* share is over distinct ICU stays. *Stays ever positive* equals the row-level rate for T3 because the task is one-per-stay. *SOFA hosp. max* is the worst SOFA score across the full hospitalization [*admission_dttm*, *discharge_dttm*], computed via `clifpy`'s polars backend (`compute_sofa_polars`).

Cohort parquet schema. Ten columns: `hospitalization_id`, `window_start`, `window_end`, `<label>`, `split`, `age_at_admission`, `sex_category`, `race_category`, `ethnicity_category`, and the task-specific `label_*` column.

Federated protocol

- Two-layer privacy model: (i) CLIF-WorkBench Singularity container — no internet, no root, immutable read-only image; (ii) aggregate-only outputs from `flair build-report`.
- Library-enforced HIPAA safeguards: cell suppression in `table1.json` for counts <10; subgroup drop in `fairness.json` for subgroups with $n < 10$; calibration-bin suppression in `curves.json`.
- Audit chain: SHA-256 of each task script is stamped into every output JSON; the central aggregator verifies byte-identical task code across sites.
- Submission contract: user appends `y_pred`/`y_prob` to the cohort parquet; `flair build-report` emits exactly four JSON files. No patient-level data ever leaves the site. For the FLAIR baseline kit, the container additionally ships the tokenizer and frozen weights; sites invoke `flair predict-baseline <task>` instead of writing their own feature/model code.

Baseline kit

Generic event tokenizer (count featurization). The benchmark is set by a count-featurization baseline following the EHRSHOT/FEMR recipe (Wornow et al., 2023; Steinberg et al., 2021), adapted to CLIF. Each row in a CLIF event table (`vitals`, `labs`, `medication_admin_intermittent`, `medication_admin_continuous`, `respiratory_support`) is mapped to a deterministic token of the form `<table>.<category>` (e.g., `vitals.heart_rate`, `labs.creatinine`, `meds.norepinephrine`). For each cohort window, FLAIR emits a sparse token → integer-count vector counting events in [`window_start`, `window_end`]. The tokenizer is config-free and exhaustive — every category present in the input flows through with no per-task curation. Because tokens are derived from CLIF schema-locked category fields, the same code runs unchanged on MIMIC-CLIF and consortium sites. Unlike EHRSHOT, which counts ontology-expanded source codes over the entire record via an Athena

ontology, FLAIR counts schema-locked CLIF category tokens inside the task's cohort window, dropping the external-ontology dependency. Module: `flair_benchmark/baseline/tokenizer.py`.

Logistic-regression baseline. Sparse count features → L2-penalized logistic regression (`sklearn`); submitted as `y_prob` for the binary tasks.

XGBoost baseline. The same sparse count features → XGBoost classifier with default depth and learning rate; submitted as `y_prob`.

Both models are trained on MIMIC-CLIF (70/30 temporal split via `_split.py`) for each of T1–T5, serialized into the CLIF-WorkBench Singularity container, and shipped to each consortium site. Hyperparameters, the token dictionary, and the missingness convention (absent token = zero count) are deferred to the Supplement (Model Cards + Datasheet).

Reporting and statistical analysis

- TRIPOD+AI (Collins et al., 2024) compliance via `flair_benchmark/report/tripod_ai.py` auto-population.
- Primary metrics: AUROC, AUPRC; 95% CIs via 1000-iteration bootstrap.
- Calibration: slope, intercept, Brier score, reliability curves (downsampled raw points in `curves.json`).
- Fairness: per-subgroup AUROC and parity gaps across sex, race, ethnicity, and age bins.
- Central dashboard renders ROC / PR / calibration curves from the aggregated `curves.json` without re-accessing patient data.

3 Results *[≈ 700 words]*

Cohort

- **Table 1.** Combined cohort characteristics across N sites (HIPAA-suppressed cells where applicable), stratified by split and label.
- **Figure 2 callout:** CONSORT-style flow diagram per task (raw → cohort filters → train/test). Mirrors EchoNext-Mini Fig. 2 layout.
- Reported counts: total hospitalizations, total ICU stays per task, per-task prevalence, demographic distribution.

Baseline performance on MIMIC-CLIF

- Per-task internal test AUROC / AUPRC with 95% bootstrap CIs, reported separately for each model family (logistic regression, XGBoost).
- Calibration slope / intercept / Brier score per task per model family.

Federated performance across the CLIF consortium

- **Table 2.** Per-task site-level summary (median, IQR, range for AUROC / AUPRC across N sites), with a `model_family` column so each task carries one row per family per site.
- **Figure 3.** Forest plot of per-site AUROC by task with two glyphs per task (logistic regression vs XGBoost); vertical reference line at the MIMIC-CLIF internal benchmark for that family.
- Quantify calibration drift between MIMIC-CLIF and held-out consortium sites, separately by model family.

Fairness

- **Figure 4.** Heatmap of per-subgroup AUROC (rows = tasks, columns = subgroups) with parity-gap shading. Cells with $n < 10$ are suppressed.

Reproducibility evidence

- Bit-identical SHA-256 task-script hashes recorded in every output JSON across sites; the central aggregator's audit log confirms protocol fidelity.

4 Discussion *[≈ 400 words]*

- **Generalization.** How much do MIMIC-trained baselines drop when evaluated on held-out consortium sites? Quantify and contextualize.
- **Positioning vs. prior benchmarks.** YAIB (van de Water et al., 2023) and HiRID-ICU-Benchmark (Yèche et al., 2021) harmonize public sources only; FLAIR's differentiator is federation across private data with an audit chain.
- **Limitations.** `clifpy` data-quality dependency; T5 CKD-stage exclusion not yet implemented (CLIF `Diagnoses` table not exposed in FLAIR loaders); T4 antibiotic drug-list curation is an open dependency; baseline tokens are pure counts within the prediction window with no time-of-occurrence, magnitude, or trend features (deliberate floor; richer encodings are left to user submissions); no urine-output criterion for AKI.
- **Bias and fairness.** Consortium sites self-select; rural and safety-net hospitals are under-represented.
- **Intended use.** External validation tool for development and method comparison — *not* a leaderboard endorsing production deployment.
- **Regulatory framing.** Per FDA CDS guidance and IMDRF SaMD clinical-evaluation guidance (see `docs/paper/lit/`), the benchmark targets the "inform clinical management" use class, not autonomous decision-making.

5 Conclusions *[≈ 120 words]*

- One paragraph. FLAIR delivers a federated, schema-locked, audit-traceable ICU benchmark **paired with a reusable baseline kit (event tokenizer + logistic regression + XGBoost) that defines the floor of expected performance at every site**; full stack released open source (Apache 2.0, `github.com/clif-consortium/FLAIR`) — code, container, baseline weights, and per-task LaTeX schematics. Researchers and clinicians are invited to grow the catalog by submitting new tasks as single-file Python modules via GitHub PR, accepted through a two-site governance review.

Required statements

- **Data and Code Availability.** GitHub `clif-consortium/FLAIR` (Apache 2.0); MIMIC-CLIF on PhysioNet; baseline weights on Zenodo **TODO:** DOI placeholder; Singularity image tag on GHCR **TODO:** tag placeholder.
- **IRB.** Each consortium site obtained independent IRB approval; central protocol coordination at the University of Chicago **TODO:** confirm.
- **Funding.** **TODO:** grants.
- **Author Contributions.** CRediT taxonomy slots **TODO:** fill.
- **Conflicts of Interest.** **TODO:** disclosures.
- **AI-Use Disclosure.** **TODO:** which LLM helped where.

Supplementary Appendix *[no NEJM AI hard cap]*

- Per-task LaTeX schematics (T1–T5), compiled from `docs/task_schematics/`; the sepsis-T4 schematic (Kamran et al., 2024) already exists and is the template.
- Datasheet for Datasets (Gebru-style).
- Model Cards (Mitchell-style) for the two baseline models (logistic regression and XGBoost).
- Tokenizer specification — deterministic `<table>.<category>` mapping; vocabulary size on MIMIC-CLIF; missingness convention (absent token = zero count); per-token frequency histograms.
- Full TRIPOD+AI checklist.
- Hyperparameter table, feature dictionary, missingness table.
- Per-site Table-1 details (cells with $n \geq 10$).

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